

STAT 423 Winter 2026 - Final Project

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2026-03-08

Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
  orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
  ifelse(grepl("Country", orig_data$cdr_genre), "Country",
  ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
  ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
  danceability = orig_data$danceability, happiness = orig_data$happiness,
  loudness_db = orig_data$loudness_d_b,
  length_sec = orig_data$length_sec)

cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
  unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- subpop_60s <- cleaned_data[cleaned_data$year >= i - 10 &
    cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
```

```

if(nrow(subpop) < 50) {
  message(paste("Skipping", decade_label, "- Not enough data points."))
  next
}

factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
  function(x) length(unique(na.omit(x))))

if(any(factor_counts < 2)) {
  message(paste("Skipping", decade_label, "due to single-level factor:",
    names(factor_counts)[factor_counts < 2]))
  next
}

first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
  + danceability + happiness + bpm + loudness_db + length_sec)

second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
  + age*energy + age*danceability + age*happiness + age*bpm
  + age*loudness_db + age*length_sec + race*genre + race*energy
  + race*danceability + race*happiness + race*bpm + race*loudness_db
  + race*length_sec + gender*genre + gender*energy + gender*danceability
  + gender*happiness + gender*bpm + gender*loudness_db + gender*length_sec)

if(anova(first_model, second_model)$"Pr(>F)"[2] <= 0.1){
  optimal_model <- step(second_model, direction="backward")
} else {
  optimal_model <- step(first_model, direction="backward")
}

model_results[[decade_label]] <- list(
  "first" = first_model,
  "second" = second_model,
  "optimal" = optimal_model
)

print(decade_label)

summary(first_model)

summary(second_model)

summary(optimal_model)
}

```

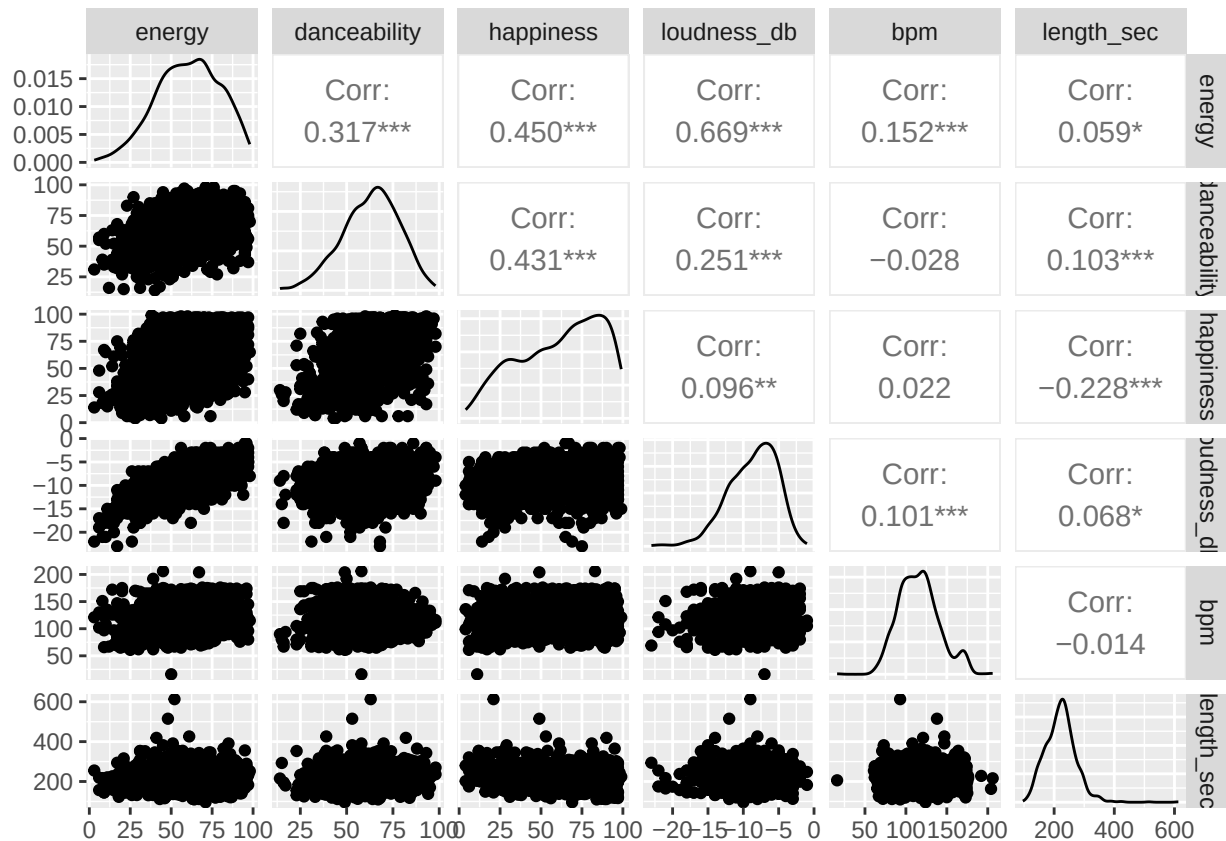
```
## Skipping 1950s - Not enough data points.
```

```
## Skipping 2020s due to single-level factor: genre
```

```

library("GGally")
ggpairs(data = cleaned_data[,c("energy", "danceability", "happiness", "loudness_db", "bpm", "length_sec

```



Loop through each decade in the model_results list

```
for (decade in names(model_results)) {
```

Check if the 'optimal' model exists for this decade

```
if (!is.null(model_results[[decade]]$optimal)) {
```

```
  cat("\n", "=====", "\n")
```

```
  cat(" SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
```

```
  cat("=====", "\n")
```

Print the summary

```
  print(summary(model_results[[decade]]$optimal))
```

```
} else {
```

```
  cat("\nNo optimal model found for:", decade, "\n")
```

```
}
```

```
}
```

```
##
```

```
## =====
```

```
## SUMMARY FOR OPTIMAL MODEL: 1960s
```

```
## =====
```

```
##
```

```
## Call:
```

```
## lm(formula = rating ~ age + gender + race + genre + energy +
```

```
## danceability + happiness + length_sec + race:genre + race:energy +
```

```
## race:danceability + gender:danceability + gender:happiness,
```

```
## data = subpop)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.235892   2.001360   3.116  0.00214 **
## age           -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male -1.820687   1.293523  -1.408  0.16101
## raceWhite      1.604179   1.795663   0.893  0.37287
## genreJazz      1.473039   1.332354   1.106  0.27040
## genreOther     1.978211   0.999147   1.980  0.04926 *
## genrePop       0.988531   0.889417   1.111  0.26788
## genreRock      0.260014   0.629236   0.413  0.67994
## energy        -0.005765   0.015336  -0.376  0.70743
## danceability   -0.013030   0.022465  -0.580  0.56264
## happiness      0.004845   0.009013   0.538  0.59154
## length_sec     0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz -1.353373  2.222521  -0.609  0.54334
## raceWhite:genreOther -3.249204  1.187123  -2.737  0.00683 **
## raceWhite:genrePop -1.786329  0.784348  -2.277  0.02395 *
## raceWhite:genreRock      NA         NA      NA      NA
## raceWhite:energy    0.024399  0.017347   1.407  0.16132
## raceWhite:danceability -0.046982  0.022464  -2.091  0.03791 *
## genderNot Male:danceability 0.076584  0.023552   3.252  0.00137 **
## genderNot Male:happiness -0.030694  0.016106  -1.906  0.05830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314, Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF, p-value: 4.551e-08
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1970s
## =====
##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1615 -1.0209  0.0511  1.0964  5.0519
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.250133   0.992268   5.291 2.77e-07 ***
## age           -0.033100   0.020798  -1.591  0.112830
## raceWhite     -0.430183   0.299921  -1.434  0.152800
## genreJazz     -0.547224   1.924576  -0.284  0.776402
## genreOther     0.979427   0.660001   1.484  0.139144
## genrePop      -0.607790   0.641502  -0.947  0.344375
```

```

## genreRock    0.182210    0.658709    0.277 0.782315
## length_sec   0.009494    0.002515    3.775 0.000202 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.808 on 237 degrees of freedom
## Multiple R-squared:  0.2243, Adjusted R-squared:  0.2013
## F-statistic: 9.787 on 7 and 237 DF,  p-value: 1.045e-10
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1980s
## =====
##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2377 -1.0513 -0.0178  0.9701  4.2309
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.960976    0.549218   9.033  <2e-16 ***
## energy         0.009412    0.005806   1.621   0.1064
## danceability   0.015115    0.009255   1.633   0.1038
## happiness     -0.011560    0.005495  -2.104   0.0365 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.505 on 224 degrees of freedom
## Multiple R-squared:  0.02279,  Adjusted R-squared:  0.009699
## F-statistic: 1.741 on 3 and 224 DF,  p-value: 0.1594
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1990s
## =====
##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2837 -0.9790 -0.0504  0.8700  4.4145
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.465767    0.942926  10.039  < 2e-16 ***
## raceWhite     -0.469197    0.277569  -1.690   0.09337 .
## genderNot Male  0.720941    0.253791   2.841   0.00523 **
## energy        -0.035118    0.010721  -3.276   0.00135 **
## happiness      0.016936    0.006634   2.553   0.01185 *

```

```

## loudness_db      0.311544    0.066407    4.691 6.82e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.469 on 129 degrees of freedom
## Multiple R-squared:  0.2445, Adjusted R-squared:  0.2152
## F-statistic: 8.349 on 5 and 129 DF,  p-value: 7.287e-07
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2000s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    14.254799    5.689019   2.506 0.013714 *
## age            -0.262392    0.164223  -1.598 0.113013
## genderNot Male  -0.551642    2.208098  -0.250 0.803196
## raceWhite      -13.479191    3.724145  -3.619 0.000451 ***
## genreOther      0.362481    1.668695   0.217 0.828443
## genrePop        0.218854    1.608297   0.136 0.892013
## genreRock       0.090807    1.604975   0.057 0.954986
## energy          0.018882    0.011341   1.665 0.098826 .
## danceability    -0.083852    0.065397  -1.282 0.202519
## happiness        0.001354    0.011393   0.119 0.905583
## bpm            -0.014549    0.008120  -1.792 0.075985 .
## length_sec      -0.009048    0.005130  -1.764 0.080619 .
## age:danceability  0.003737    0.002195   1.703 0.091492 .
## raceWhite:genreOther -1.186109    0.866090  -1.369 0.173685
## raceWhite:genrePop      NA         NA      NA      NA
## raceWhite:genreRock     NA         NA      NA      NA
## raceWhite:danceability  0.084078    0.031491   2.670 0.008759 **
## raceWhite:happiness    -0.025028    0.018899  -1.324 0.188199
## raceWhite:bpm          0.035295    0.012426   2.840 0.005385 **
## raceWhite:length_sec    0.019703    0.011520   1.710 0.090064 .
## genderNot Male:danceability -0.041970    0.024007  -1.748 0.083262 .
## genderNot Male:happiness  0.021226    0.014487   1.465 0.145771
## genderNot Male:bpm       0.019019    0.010619   1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198

```

```
## F-statistic: 2.803 on 20 and 108 DF,  p-value: 0.0003367
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2010s
## =====
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2684 -1.0215  0.1378  0.9163  3.4440
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.788777    1.188134   4.031 0.000103 ***
## genderNot Male    0.459667    0.298784   1.538 0.126808
## energy         -0.024064    0.009966  -2.415 0.017405 *
## happiness        0.012193    0.007833   1.557 0.122440
## length_sec      0.009668    0.004602   2.101 0.037952 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.594 on 110 degrees of freedom
## Multiple R-squared:  0.09648,    Adjusted R-squared:  0.06363
## F-statistic: 2.937 on 4 and 110 DF,  p-value: 0.02378
```

Preliminary Exploration of Data through Visual Analysis

Plots of Individual Relationships

The plots show the relationship between the predictors and the response variable, rating.

```
library(ggplot2)
library(patchwork)

##
## Attaching package: 'patchwork'
##
## The following object is masked from 'package:MASS':
##
##      area

plot_age <- ggplot(cleaned_data, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

plot_energy <- ggplot(cleaned_data, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Energy")
```

```
plot_danceability <- ggplot(cleaned_data, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Danceability")
```

```
plot_happiness <- ggplot(cleaned_data, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")
```

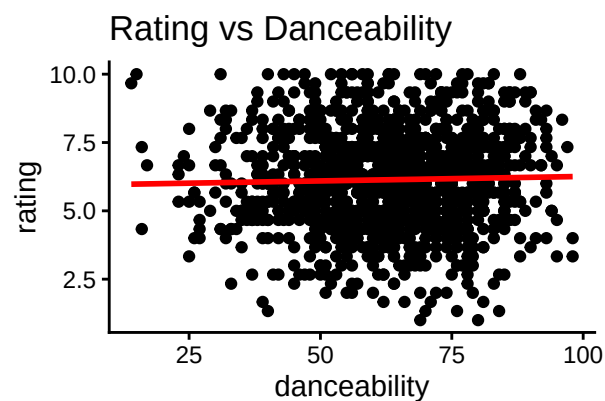
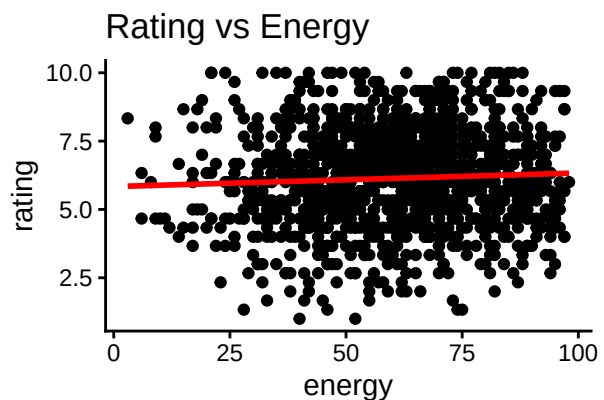
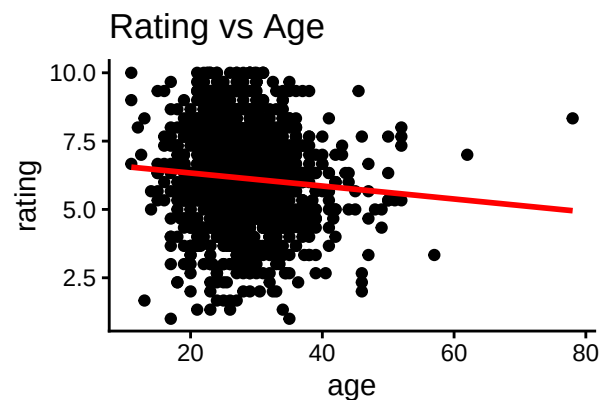
```
plot_age + plot_energy + plot_danceability + plot_happiness
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
plot_bpm <- ggplot(cleaned_data, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")
```

```
plot_loudness_db <- ggplot(cleaned_data, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
```



```

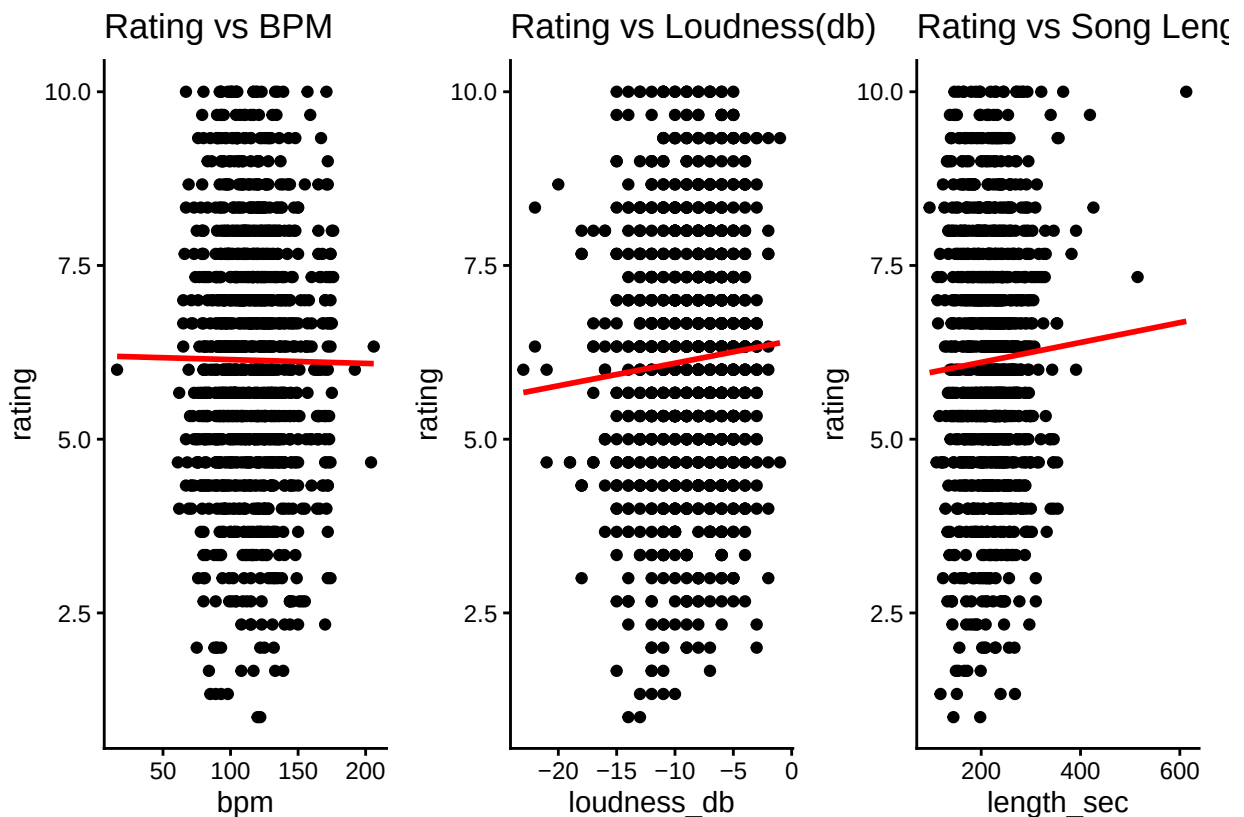
geom_smooth(method = "lm", color = "red", se = F) +
theme_classic() +
ggtitle("Rating vs Loudness(db)")

plot_length_sec <- ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length(sec)")

plot_bpm + plot_loudness_db + plot_length_sec

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



```

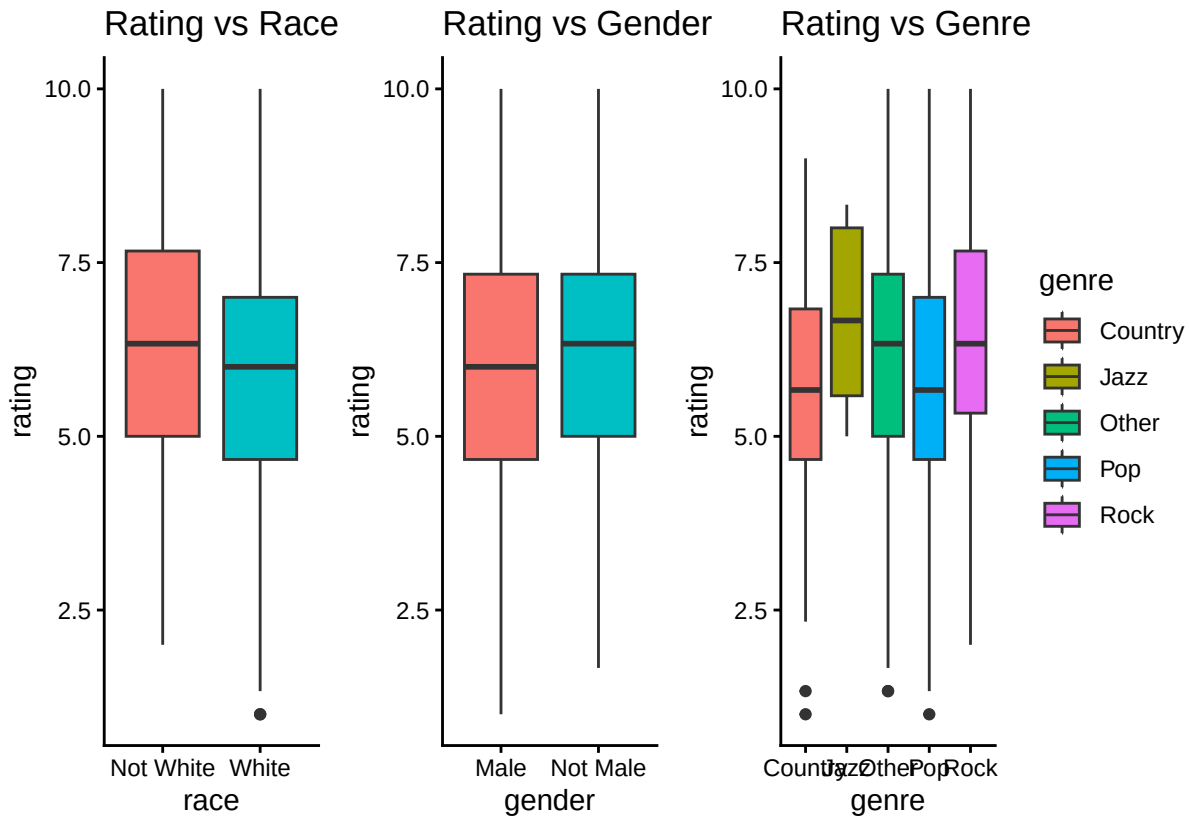
plot_race <- ggplot(cleaned_data, aes(x = race, y = rating, fill = race)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Race") +
  guides(fill = "none")

plot_gender <- ggplot(cleaned_data, aes(x = gender, y = rating, fill = gender)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Gender") +
  guides(fill = "none")

```

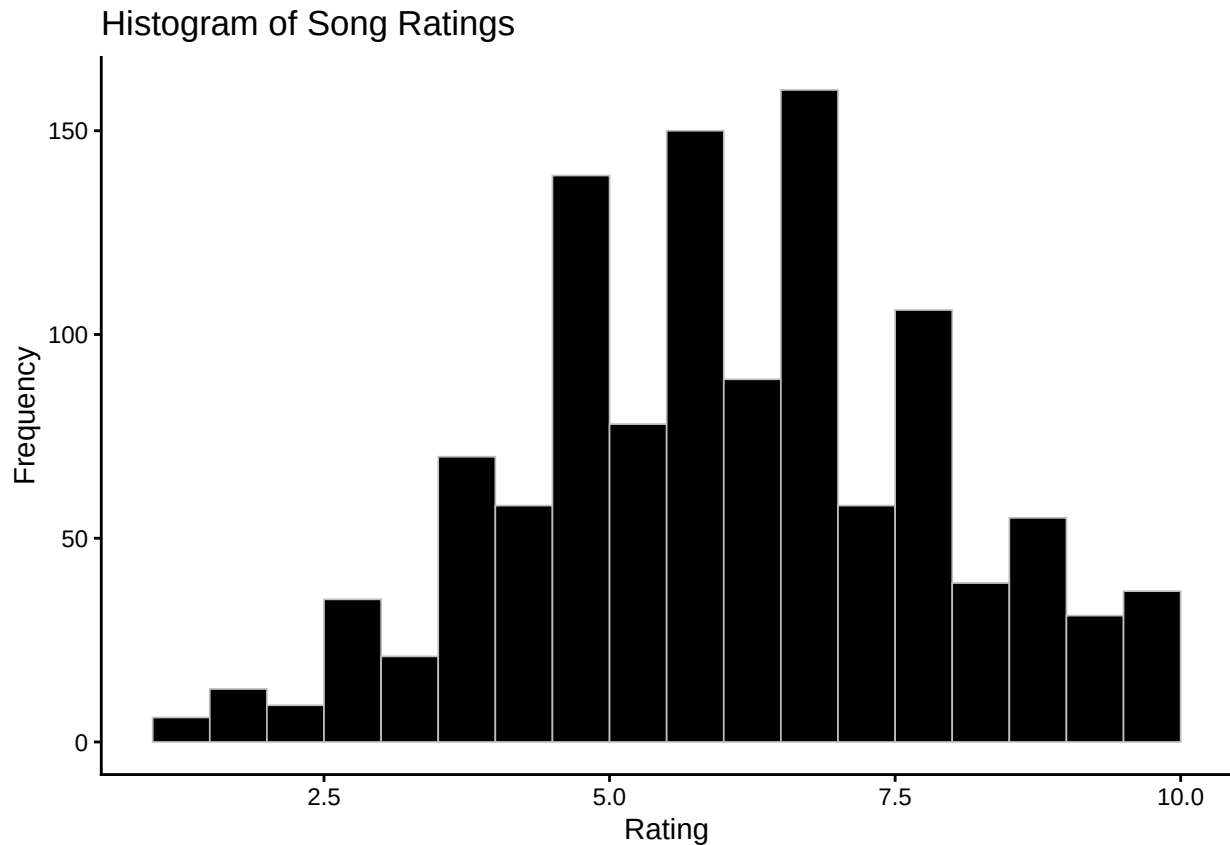
```
plot_genre <- ggplot(cleaned_data, aes(x = genre, y = rating, fill = genre)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Genre")
```

```
plot_race + plot_gender + plot_genre
```



As expected, banding is observed in the scatterplots because the response variable, rating, is discrete. Since race, gender, and genre are categorical predictors, the relationship with rating is displayed by boxplots instead.

```
ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



However, the histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. While the ratings are discrete (integer-valued) and there are many categorical predictors, the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

Plots of Grouped Relationships

Data visualizations for grouped relationships help us examine whether relationships between predictors and ratings differ across categories. These plots justify the inclusion of categorical predictors in the regression models and help identify potential interaction effects.

```
model_60s <- model_results[["1960s"]]$optimal
model_70s <- model_results[["1970s"]]$optimal
model_80s <- model_results[["1980s"]]$optimal
model_90s <- model_results[["1990s"]]$optimal
model_2000s <- model_results[["2000s"]]$optimal
model_2010s <- model_results[["2010s"]]$optimal
```

```
model_60s
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Coefficients:
```

```
##          (Intercept)                age
##          6.235892                -0.036813
##          genderNot Male                raceWhite
##          -1.820687                1.604179
##          genreJazz                genreOther
##          1.473039                1.978211
##          genrePop                genreRock
##          0.988531                0.260014
##          energy                danceability
##          -0.005765                -0.013030
##          happiness                length_sec
##          0.004845                0.008965
##          raceWhite:genreJazz                raceWhite:genreOther
##          -1.353373                -3.249204
##          raceWhite:genrePop                raceWhite:genreRock
##          -1.786329                NA
##          raceWhite:energy                raceWhite:danceability
##          0.024399                -0.046982
## genderNot Male:danceability                genderNot Male:happiness
##          0.076584                -0.030694
```

```
vars_by_decade <- lapply(model_results, function(x) {
  attr(terms(x$optimal), "term.labels")
})

all_vars <- unique(unlist(vars_by_decade))

presence_matrix <- sapply(vars_by_decade, function(vars) {
  all_vars %in% vars
})

rownames(presence_matrix) <- all_vars
presence_matrix
```

```
##          1960s 1970s 1980s 1990s 2000s 2010s
## age          TRUE  TRUE FALSE FALSE  TRUE FALSE
## gender        TRUE FALSE FALSE  TRUE  TRUE  TRUE
## race          TRUE  TRUE FALSE  TRUE  TRUE FALSE
## genre         TRUE  TRUE FALSE FALSE  TRUE FALSE
## energy        TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## danceability  TRUE FALSE  TRUE FALSE  TRUE FALSE
## happiness     TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## length_sec    TRUE  TRUE FALSE FALSE  TRUE  TRUE
## race:genre    TRUE FALSE FALSE FALSE  TRUE FALSE
## race:energy   TRUE FALSE FALSE FALSE FALSE FALSE
## race:danceability TRUE FALSE FALSE FALSE  TRUE FALSE
## gender:danceability TRUE FALSE FALSE FALSE  TRUE FALSE
## gender:happiness TRUE FALSE FALSE FALSE  TRUE FALSE
## loudness_db   FALSE FALSE FALSE  TRUE FALSE FALSE
## bpm           FALSE FALSE FALSE FALSE  TRUE FALSE
## age:danceability FALSE FALSE FALSE FALSE  TRUE FALSE
## race:happiness FALSE FALSE FALSE FALSE  TRUE FALSE
## race:bpm      FALSE FALSE FALSE FALSE  TRUE FALSE
## race:length_sec FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:bpm     FALSE FALSE FALSE FALSE  TRUE FALSE
```

```
rownames(presence_matrix)[rowSums(presence_matrix) == ncol(presence_matrix)]
```

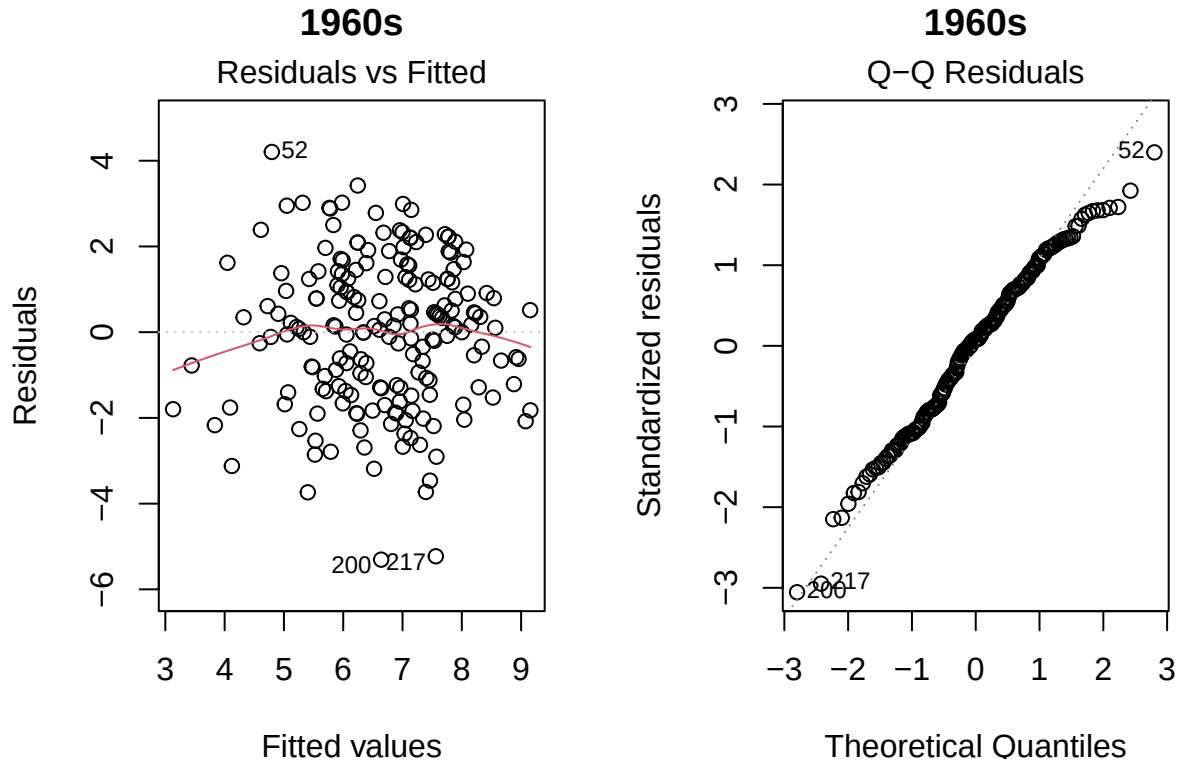
```
## character(0)
```

```
par(mfrow = c(1,2))
```

```
plot(model_60s, which = 1:2, main = "1960s")
```

```
## Warning: not plotting observations with leverage one:
```

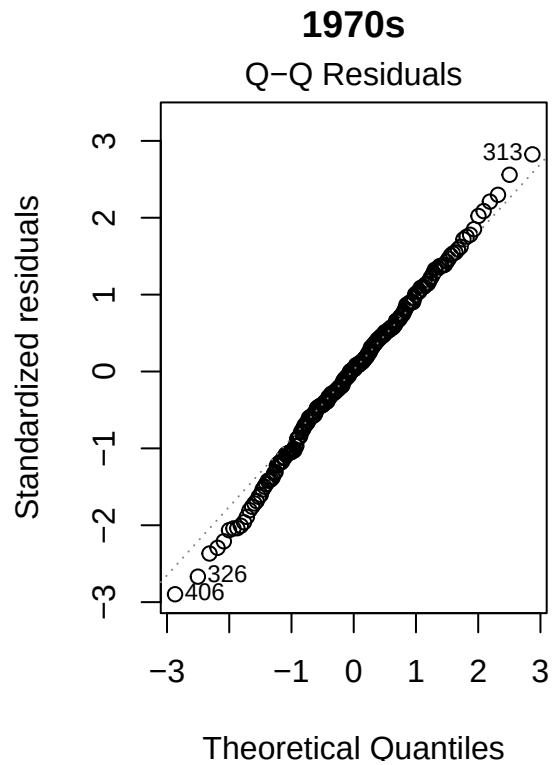
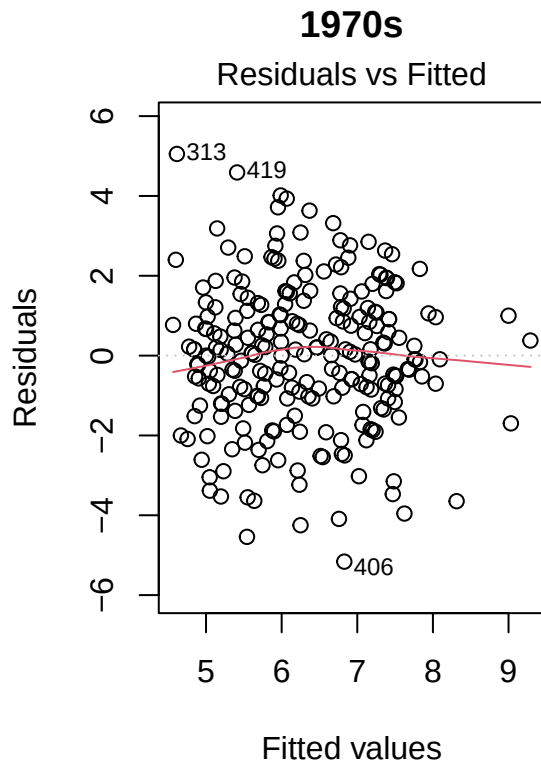
```
## 48
```



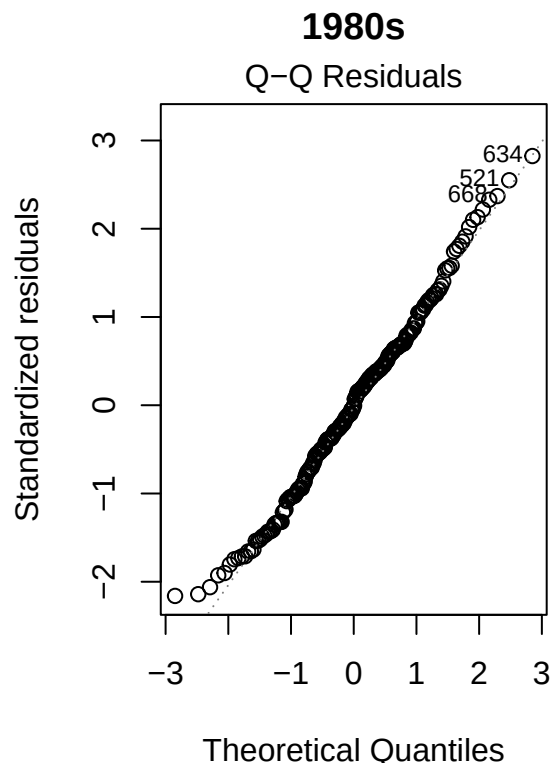
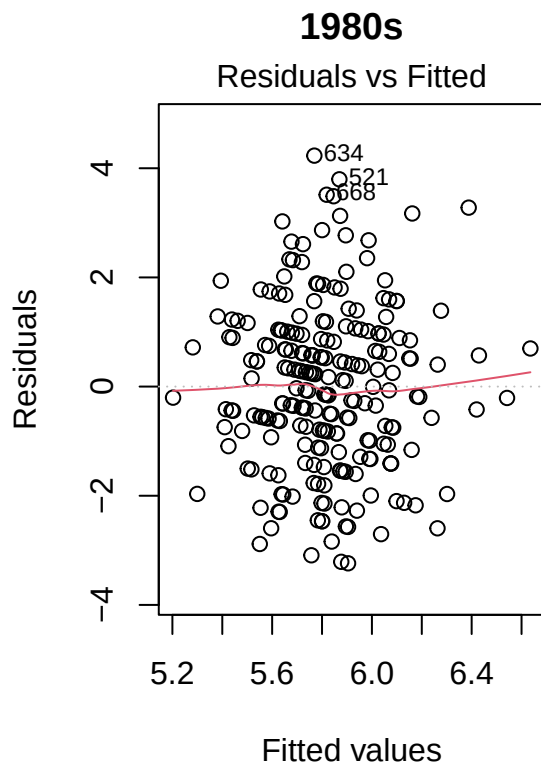
```
plot(model_70s, which = 1:2, main = "1970s")
```

```
## Warning: not plotting observations with leverage one:
```

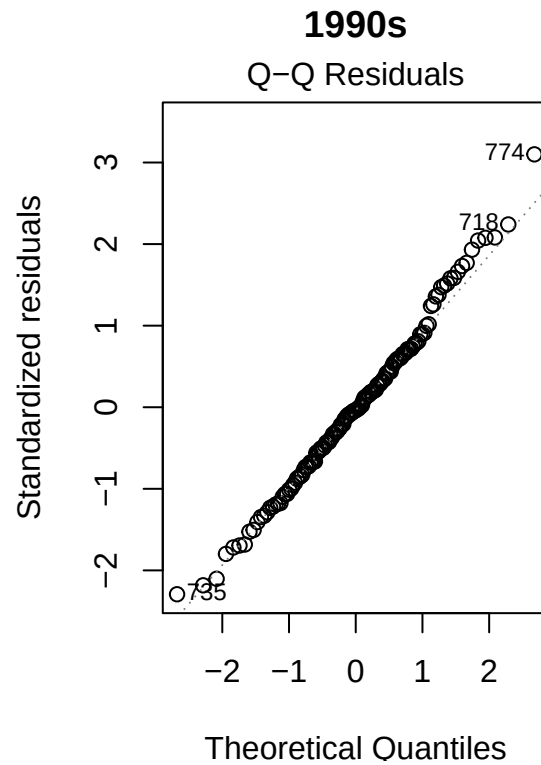
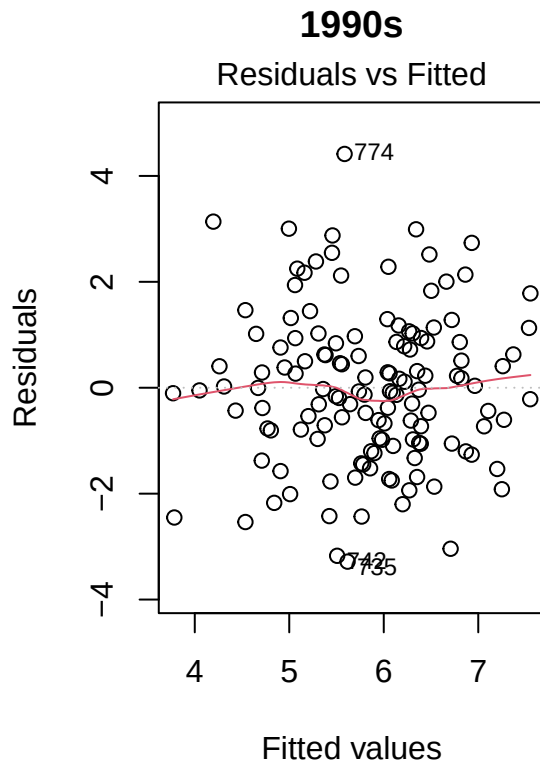
```
## 239
```



```
plot(model_80s, which = 1:2, main = "1980s")
```

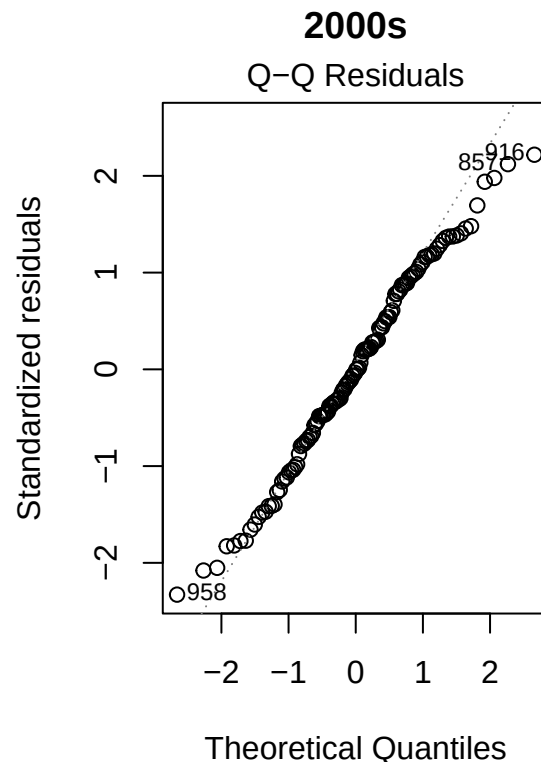
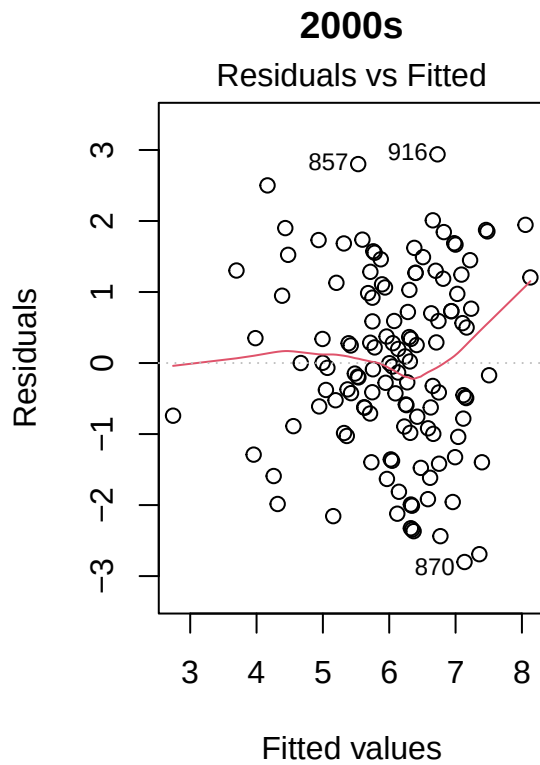


```
plot(model_90s, which = 1:2, main = "1990s")
```

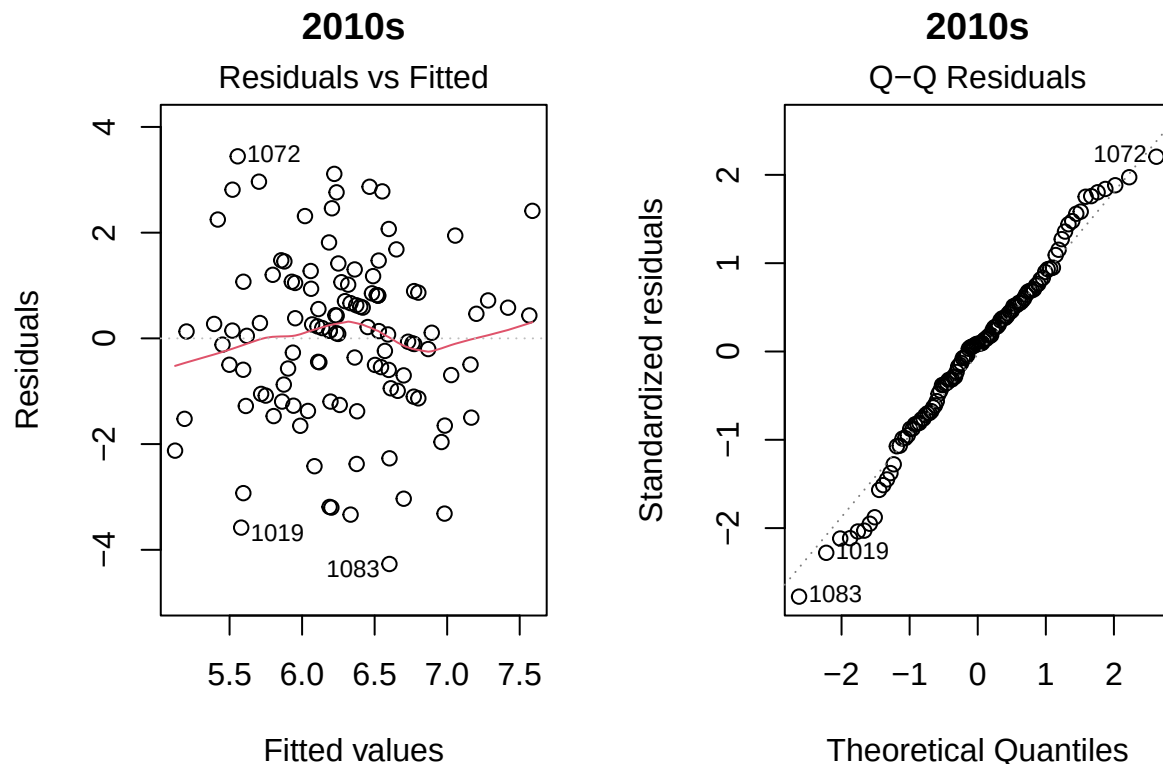


```
plot(model_2000s, which = 1:2, main = "2000s")
```

```
## Warning: not plotting observations with leverage one:
## 4
```

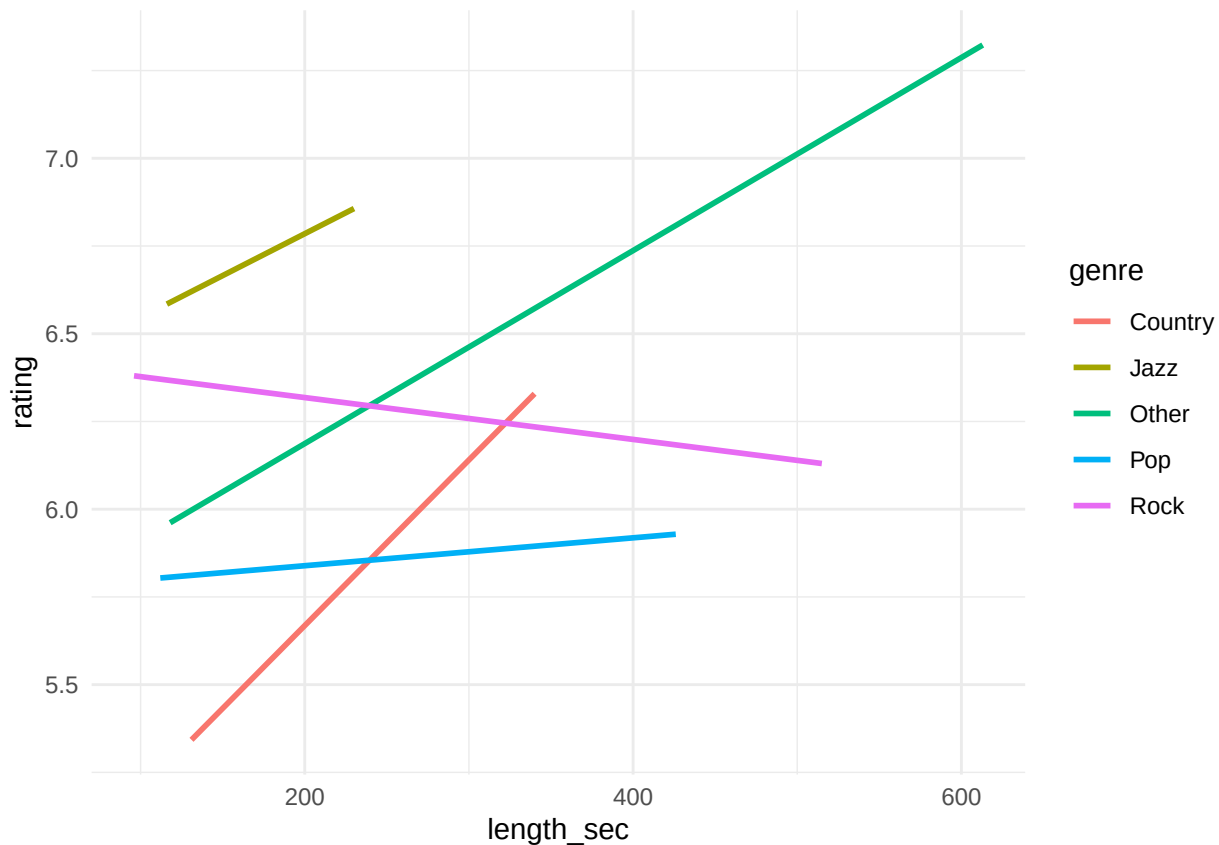


```
plot(model_2010s, which = 1:2, main = "2010s")
```



```
ggplot(cleaned_data, aes(x = length_sec, y = rating)) +  
  geom_smooth(aes(color = genre), method = "lm", se = FALSE) +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

The fitted regression lines indicate that the relationship between song length and rating varies by genre. In particular, certain genres (e.g., Other and Country) exhibit steeper positive slopes, while others (e.g., Rock and Pop) show weaker or slightly negative trends, suggesting a potential interaction between length and genre.

```
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.4.3
```

```
coef_1960s <- tidy(model_60s, conf.int = TRUE) %>% mutate(decade = "1960s")
coef_1970s <- tidy(model_70s, conf.int = TRUE) %>% mutate(decade = "1970s")
coef_1980s <- tidy(model_80s, conf.int = TRUE) %>% mutate(decade = "1980s")
coef_1990s <- tidy(model_90s, conf.int = TRUE) %>% mutate(decade = "1990s")
coef_2000s <- tidy(model_2000s, conf.int = TRUE) %>% mutate(decade = "2000s")
coef_2010s <- tidy(model_2010s, conf.int = TRUE) %>% mutate(decade = "2010s")
```

```
coef_data <- bind_rows(
  coef_1960s,
  coef_1970s,
  coef_1980s,
  coef_1990s,
  coef_2000s,
  coef_2010s
)
```

```
coef_data <- coef_data %>%
  filter(term != "(Intercept)") %>%
  filter(grepl("^genre", term)) %>%
```

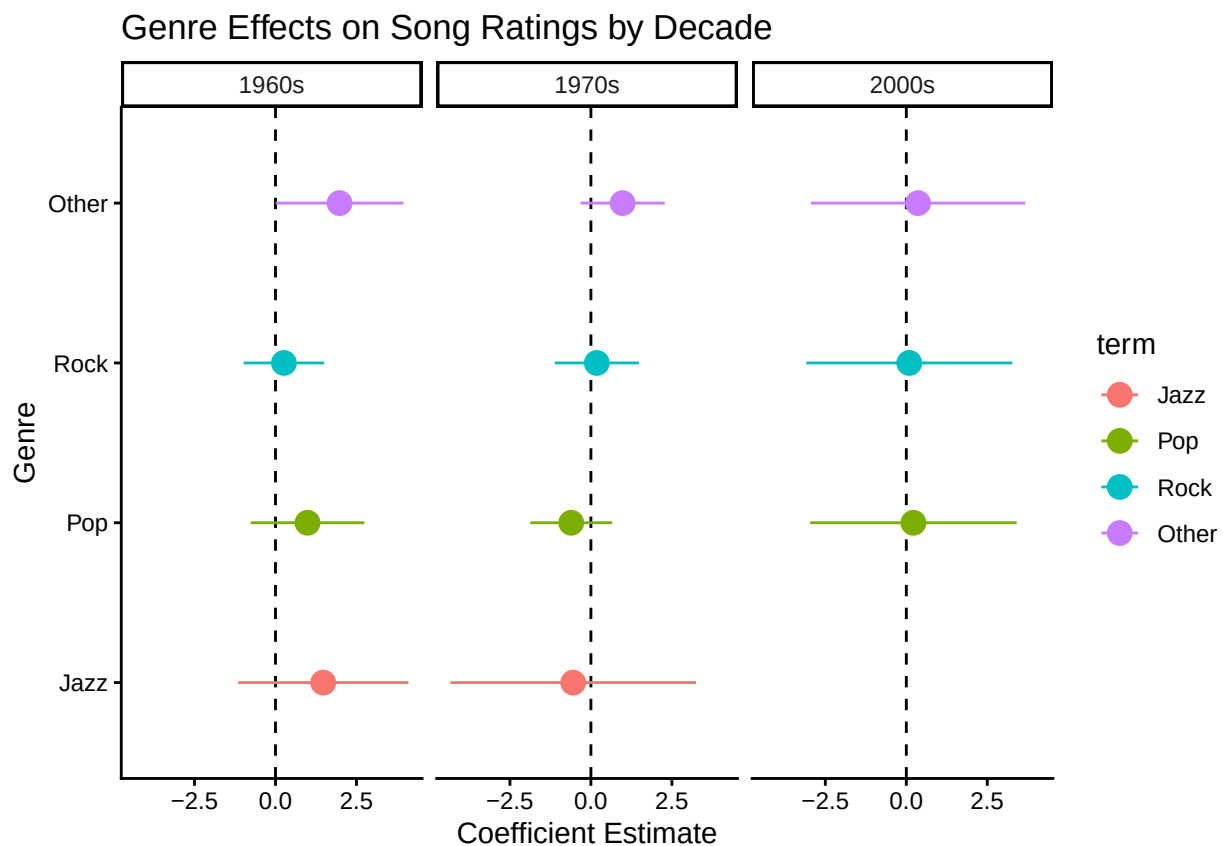
```

mutate(term = gsub("^genre", "", term))

coef_data$term <- factor(
  coef_data$term,
  levels = c("Jazz", "Pop", "Rock", "Country", "HipHop", "Other")
)

ggplot(coef_data, aes(x = estimate, y = term, color = term)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_pointrange(aes(xmin = conf.low, xmax = conf.high), size = 0.8) +
  facet_wrap(~ decade) +
  theme_classic() +
  labs(
    title = "Genre Effects on Song Ratings by Decade",
    x = "Coefficient Estimate",
    y = "Genre"
  )

```



```
model_60s
```

```

##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##

```

```

## Coefficients:
##          (Intercept)                age
##          6.235892          -0.036813
##          genderNot Male          raceWhite
##          -1.820687          1.604179
##          genreJazz          genreOther
##          1.473039          1.978211
##          genrePop          genreRock
##          0.988531          0.260014
##          energy          danceability
##          -0.005765          -0.013030
##          happiness          length_sec
##          0.004845          0.008965
##          raceWhite:genreJazz          raceWhite:genreOther
##          -1.353373          -3.249204
##          raceWhite:genrePop          raceWhite:genreRock
##          -1.786329          NA
##          raceWhite:energy          raceWhite:danceability
##          0.024399          -0.046982
## genderNot Male:danceability          genderNot Male:happiness
##          0.076584          -0.030694

model_70s

##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Coefficients:
## (Intercept)          age          raceWhite          genreJazz          genreOther          genrePop
##    5.250133   -0.033100   -0.430183   -0.547224    0.979427   -0.607790
## genreRock    length_sec
##    0.182210    0.009494

model_80s

##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Coefficients:
## (Intercept)          energy          danceability          happiness
##    4.960976    0.009412    0.015115   -0.011560

model_90s

##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Coefficients:
## (Intercept)          raceWhite          genderNot Male          energy          happiness
##    9.46577    -0.46920    0.72094   -0.03512    0.01694
## loudness_db
##    0.31154

```

```
model_2000s
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Coefficients:
##              (Intercept)              age
##              14.254799             -0.262392
##            genderNot Male             raceWhite
##             -0.551642             -13.479191
##            genreOther              genrePop
##             0.362481              0.218854
##            genreRock              energy
##             0.090807              0.018882
##            danceability             happiness
##             -0.083852              0.001354
##              bpm              length_sec
##             -0.014549             -0.009048
##            age:danceability    raceWhite:genreOther
##             0.003737              -1.186109
##            raceWhite:genrePop    raceWhite:genreRock
##              NA              NA
##            raceWhite:danceability    raceWhite:happiness
##             0.084078             -0.025028
##            raceWhite:bpm    raceWhite:length_sec
##             0.035295              0.019703
## genderNot Male:danceability    genderNot Male:happiness
##             -0.041970              0.021226
##            genderNot Male:bpm
##             0.019019
```

```
model_2010s
```

```
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Coefficients:
##      (Intercept)  genderNot Male      energy      happiness      length_sec
##      4.788777      0.459667      -0.024064      0.012193      0.009668
```

```
anova(model_results[["1960s"]]$first, model_results[["1960s"]]$second)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##           happiness + bpm + loudness_db + length_sec
```

```
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##           danceability + age * happiness + age * bpm + age * loudness_db +
```

```

##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      183 673.25
## 2      155 535.98 28      137.27 1.4178 0.09454 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_results[["1970s"]] $\$$ first, model_results[["1970s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      231 764.63
## 2      204 666.81 27      97.812 1.1083 0.3327

anova(model_results[["1980s"]] $\$$ first, model_results[["1980s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      216 494.88
## 2      192 459.91 24      34.968 0.6083 0.9247

anova(model_results[["1990s"]] $\$$ first, model_results[["1990s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *

```

```

##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      122 269.95
## 2       98 217.50 24     52.448 0.9846 0.493
anova(model_results[["2000s"]] $\$$ first, model_2000s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + length_sec + age:danceability + race:genre +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:danceability + gender:happiness + gender:bpm
## Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      116 270.33
## 2      108 208.61 8     61.719 3.994 0.0003515 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
anova(model_2000s, model_results[["2000s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + length_sec + age:danceability + race:genre +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:danceability + gender:happiness + gender:bpm
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      108 208.61
## 2       93 194.20 15     14.415 0.4602 0.9544
anova(model_results[["2010s"]] $\$$ first, model_2010s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ gender + energy + happiness + length_sec
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      103 274.94
## 2      110 279.47 -7     -4.5351 0.2427 0.9735
anova(model_2010s, model_results[["2010s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ gender + energy + happiness + length_sec

```

```

## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1    110 279.47
## 2     81 206.34 29    73.132 0.9899 0.4938

```